

1. Administrative & Study Identification

Full Study Title: A Multicenter, Open-Label, Extension Study Evaluating the Safety and Efficacy of Bomedemstat for the Treatment of Participants Enrolled in a Prior Bomedemstat Clinical Study **Short Title/Acronym:** Shorespan-017 **Protocol Number:** MK-3543-017 (PR000116787) **NCT Number:** NCT06351631 **Sponsor Name:** Merck Sharp & Dohme LLC (Imago BioSciences, Inc.) **Investigational Product:** Bomedemstat (MK-3543 / IMG-7289) **Phase of Development:** Phase III **Medical Monitor:** [Name and Contact Information]

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To transition participants into an extension study to evaluate the long-term safety and tolerability of bomedemstat. **Secondary Objectives:** To assess long-term efficacy, durability of hematologic remission, durability of response, and the incidence of disease progression (transformation to myelofibrosis, myelodysplastic syndrome, or acute myeloid leukemia).

2.2 Background & Rationale Myeloproliferative neoplasms (MPNs), such as essential thrombocythemia (ET), polycythemia vera (PV), and myelofibrosis (MF), cause vascular events and systemic symptoms sustained by platelet abnormalities and cytokine imbalance. Bomedemstat is an irreversible inhibitor of lysine-specific demethylase 1 (LSD1), an epigenetic modifier involved in regulating megakaryocyte and erythrocyte maturation. While bomedemstat has shown manageable safety and promising clinical activity, long-term evaluation is needed. This extension study is being conducted to assess the sustained safety and efficacy of bomedemstat in participants who received prior treatment with the drug and derived clinical benefit in previous feeder studies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Single Group Assignment) **Blinding:** Open-label **Randomization:** Nonrandomized (N/A)

3.2 Planned Enrollment & Duration Target \$N\$: Dependent on eligible rollover participants from feeder studies (e.g., ~56+ participants based on early cutoffs). **Number of Sites:** Multicenter (Global, including sites in the US, Italy, Australia,

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults ≥ 18 years of age. 2. Diagnosed with a myeloproliferative neoplasm (ET, PV, or MF). 3. Transitioned from a previous bomedemstat feeder study (e.g., IMG-7289-202/MK-3543-005) sponsored by Imago/Merck and established as MK-3543-017 ready. 4. Received at least 6 months of treatment with bomedemstat in the feeder study, safely tolerated the drug, and received clinical benefit in the estimation of the investigator. (For ET/PV participants from studies other than 202/005, confirmed hematologic remission must have been achieved).

4.2 Exclusion Criteria 1. History of noncompliance in a prior bomedemstat study. 2. Pregnant or breastfeeding females. 3. Participants who progressed to post-ET/PV myelofibrosis, myelodysplastic syndrome, or acute myeloid leukemia during the feeder study.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Bomedemstat 10, 15, 20, or 50 mg oral capsules taken once daily. Participants initiate treatment at the same dose that was determined safe and efficacious at the end of their feeder study. **Route of Administration:** Oral **Duration of Treatment:** Up to 10 years, as long as the participant derives clinical benefit.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for MPN-related symptoms as deemed appropriate by the investigator. **Prohibited:** Concurrent participation in another investigational therapeutic study, or transition to alternative cytoreductive therapies conflicting with bomedemstat.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening	Feeder Study End, Eligibility confirmation, Informed Consent, Baseline labs
2	Baseline	Day 0
3	Interim	Transition to extension study, IP dispensing
4	Interim	Every 4 Weeks
5	Interim	Clinic visit, Safety Labs, Efficacy Assessment, IP Accountability
6	End of Study	Up to 10 Years
7	End of Study	Final Vital Signs, Exit Interview, Disease progression assessment

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Long-term safety and tolerability, measured by the incidence and severity of treatment-emergent adverse events (TEAEs), serious adverse events (SAEs), and study discontinuations due to AEs over the study duration. **Measurement Method:** Clinical assessments and laboratory monitoring graded per standard criteria.

7.2 Secondary & Exploratory Endpoints - Duration of hematologic remission. - Percentage of participants with transformation to post-ET/PV MF, myelodysplastic syndrome (MDS), or acute myeloid leukemia (AML). - Spleen volume changes assessed by magnetic resonance imaging (MRI) or computed tomography (CT).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during the 4-weekly clinic visits (tracking common AEs such as diarrhea, contusion, abdominal pain, and upper respiratory tract infections). **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline to the sponsor. **Data Monitoring Committee (DMC):** Safety review committee overseeing the extension data globally.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who transition and receive at least one dose of bomedemstat in the extension phase). **Sample Size Justification:** No hypothesis testing will be conducted in this study. The sample size is strictly based on the pool of eligible rollover participants from previous bomedemstat clinical trials. **Handling of Missing Data:** Descriptive statistics will be utilized to summarize long-term safety and efficacy follow-up; data will be censored at the time of study discontinuation.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard site monitoring to ensure protocol adherence during the extended 4-weekly visits. **Audit Trail:** eCRF locking upon data cut-offs (e.g., annual safety updates) and routine data clarification processes.

11. Study Identifiers & Governance

Primary ID: MK-3543-017 **Registry Number:** NCT06351631 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** A Study to Evaluate Safety and Efficacy of Bomedemstat (MK-3543-017)
Official Regulatory Title: A Multicenter, Open-Label, Extension Study Evaluating the Safety and Efficacy of Bomedemstat for the Treatment of Participants Enrolled in a Prior Bomedemstat Clinical Study

12. Clinical Framework (MPN Specific)

Primary Condition: Myeloproliferative Neoplasms (ET, PV, MF)
Diagnosis Threshold: Derived clinical benefit and ≥ 6 months exposure to bomedemstat in feeder study **Medical Methodology:** Long-term LSD1 inhibitor targeted therapy **Optimization Target:** Maintenance of hematologic remission and symptom control

13. Study Procedures & Methodology

Management Pathway: Open-label extension clinical pathway
Education Protocol (HCP): Site initiation and continuous safety reporting training **Education Protocol (Patient):** At-home dosing instructions and AE self-reporting guidance **Quality Audit Mechanism:** 4-weekly clinic assessments for IP accountability and tolerability

14. Observation & Follow-Up Schedule

Total Duration: Up to 10 years (520+ weeks) **Onsite Visit Cadence:** Every 4 weeks **Remote Monitoring Frequency:** Supplemental tele-health as needed per site SOP **Checklist Review Interval:** Every 4 weeks (Q4W) during drug dispensing

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				